

Formulating Absolute Value Linear Equations - Practice

Algebra II Honors | Unit 2 | Lesson 4 Practice | TEK A2.6.D

Objective: I can formulate absolute value equations from both template and layered real-world scenarios, and classify which is which.

Section 1 — Recall

1. A machine's temperature must stay within 5% of the ideal setting of 200°F. Formulate, then identify the center and margin yourself.

Equation: _____ Center: _____

2. A runner's pace must be within 30 seconds of a target time of 360 seconds per mile. Formulate and classify: template or layered?

Equation: _____

Section 2 — Apply

3. A garden bed is symmetric about a center walkway, with rows spaced 5x feet from the walkway on each side, and total width (rows plus a 2-foot walkway) is 22 feet. Formulate, and explain why it is layered.

Equation: _____

4. A fountain sits at the center of a symmetric plaza, paving extending 3x feet from the fountain on every side, with total width (paving plus a 6-foot base) equal to 48 feet. Formulate and classify.

Equation: _____

Section 3 — Challenge

5. A conference room has chairs arranged symmetrically from a center aisle, 8x feet of width per side, with total room width (chairs plus two 4-foot doorways) equal to 96 feet. Formulate the equation.

6. The equation $2|5x| + 10 = 50$ came from a real-world scenario. Describe a possible scenario, explaining what is symmetric and what the +10 might represent.

Section 4 — Explain It

7. Explain how to tell whether a real-world scenario fits the basic $|x - \text{center}| = \text{margin}$ template or needs the layered approach instead.
